Quantum Mechanics II — Exercise Sheet 6

Spectra, resolvents and bands

Variant: Solutions

Exercise 1 (Dense eigenvalues). Let $(q_n)_{n\geq 1}$ enumerate the rationals in $[0,1]$ without repetition. On $\ell^2(\mathbb{N})$ let $(Ac)_n = q_n c_n$. Determine the spectrum and the eigenvalues. For an irrational $\lambda \in (0,1)$ construct a Weyl sequence from coordinate vectors and explain why λ is not an eigenvalue.

Solution. The operator is bounded and self-adjoint on all of ℓ^2. Every q_n is an eigenvalue with its coordinate eigenvector. If $\lambda \notin [0,1]$, the diagonal inverse has entries $(q_n-\lambda)^{-1}$ bounded by the reciprocal distance to $[0,1]$. For irrational $\lambda \in (0,1)$ choose distinct indices n_j with $|q_{n_j}-\lambda| < 1/j$. Then

$$\|e_{n_j}\| = 1, \qquad \|(A-\lambda)e_{n_j}\| = |q_{n_j}-\lambda| \longrightarrow 0.$$

There is no nonzero vector satisfying $(q_n-\lambda)c_n = 0$ for every n. The endpoints are rational eigenvalues, so altogether

$$\sigma(A) = [0,1], \qquad \sigma_{\mathrm{p}}(A) = \mathbb{Q}\cap[0,1].$$

Exercise 2 (The resolvent norm of a two-level operator). On $\mathbb{C}^2$ let $A = \operatorname{diag}(-2,3)$. For $z \in \mathbb{C}\setminus\{-2,3\}$ compute $(A-z)^{-1}$ and its norm. At $z = \mathrm{i}$ compare the exact norm with $1/|\operatorname{Im} z|$. State why this finite example does not prove equality in the lecture's resolvent bound for every self-adjoint operator.

Solution. Inverting the two diagonal entries gives

$$(A-z)^{-1} = \begin{pmatrix} (-2-z)^{-1} & 0 \\ 0 & (3-z)^{-1} \end{pmatrix}, \qquad \|(A-z)^{-1}\| = \max\{|2+z|^{-1}, |3-z|^{-1}\}.$$

At $z = \mathrm{i}$ the norm is $1/\sqrt{5}$, strictly below the bound 1. The bound is an upper estimate; its equality is not a general conclusion.

Exercise 3 (A dimerized chain). A two-site periodic chain has Bloch matrix

$$h(k) = \begin{pmatrix} 0 & \kappa_1 + \kappa_2 e^{-\mathrm{i}k} \\ \kappa_1 + \kappa_2 e^{\mathrm{i}k} & 0 \end{pmatrix}, \qquad -\pi \leq k \leq \pi,$$

where $\kappa_1, \kappa_2 > 0$. The Fourier decomposition identifies the chain Hamiltonian with multiplication by this matrix; take this identification as given. (i) Find the two bands and their gap. (ii) Treat $\kappa_1 = \kappa_2$ separately.

Solution. The characteristic equation is $E^2 = |\kappa_1 + \kappa_2 e^{\mathrm{i}k}|^2$. Thus

$$E_{\pm}(k) = \pm\sqrt{\kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos k}.$$

As $\cos k$ runs from -1 to 1, the two bands are

$$[-(\kappa_1+\kappa_2), -|\kappa_1-\kappa_2|] \quad \text{and} \quad [|\kappa_1-\kappa_2|, \kappa_1+\kappa_2].$$

The gap width is $2|\kappa_1-\kappa_2|$. For equal couplings the bands meet at zero and there is no open gap.

Exercise 4 (A resolvent calculation with a source). On $L^2(\mathbb{R})$ let A be multiplication by x^2 on its maximal domain. For $f = \mathbf{1}_{[-1,1]}$ solve $(A+1)u = f$. Verify directly that $u \in D(A)$ and compute $\|u\|^2$. Use $\int \mathrm{d}x/(1+x^2)^2 = x/[2(1+x^2)] + \arctan x/2$.

Solution. The equation is pointwise multiplication, so the candidate is

$$u(x) = \frac{\mathbf{1}_{[-1,1]}(x)}{1+x^2}.$$

Both u and x^2u are square-integrable because they are bounded and supported in $[-1,1]$. This verifies the domain as well as the action. The stated antiderivative gives

$$\|u\|^2 = \int_{-1}^{1} \frac{\mathrm{d}x}{(1+x^2)^2} = \frac{1}{2} + \frac{\pi}{4}.$$

The solution is unique because x^2+1 never vanishes.